Math 196 Problem Set 3

Due April 16, at 10:00 PM

For book problems, see the pdf on canvas or in my email.

Problem 1. Consider the vectors

$$u = \begin{bmatrix} 1/2 \\ \sqrt{3}/2 \end{bmatrix}, \; v = \begin{bmatrix} -1/2 \\ \sqrt{3}/2 \end{bmatrix}$$

Let ℓ_u be the line spanned by u, and ℓ_v be the line spanned by v.

(a) Compute the matrices representing the orthogonal projections onto ℓ_u and ℓ_v, that is proj_u and proj_v. Use these to find the projection of u onto ℓ_v and v onto ℓ_u.

(b) Compute the matrices representing reflections across ℓ_u and ℓ_v, call these S_u and S_v

(c) Show that $S_u S_v$ and $S_v S_u$ are rotation matrices. What are the angles of rotation?

Problem 2. Book Exercises 2 to 6 in Section 2.4, page 97.

Problem 3. Suppose that the quantities x_1, x_2, x_3 and y_1, y_2, y_3 are related by the equation:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

Suppose that y_1 and y_3 are held constant, and y_2 increases. How are x_1, x_2, and x_3 changing?

Problem 4. (Optional) Suppose that A is an $n \times n$ matrix. Suppose that B and C are matrices such that $AB = I_n$ and $CA = I_n$. Show that $B = C$.

Problem 5. Consider a 2×2 matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

(a) If $ad - bc = 0$, show that A is not invertible.[1]

(b) If $ad - bc \neq 0$, show using row operations that A is invertible.

(c) Compute the product

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Using this, find a formula for the inverse of a general 2×2 matrix, when it exists.

Problem 6. Let U and V be linear subspaces of $\mathbb{R}^n$.

(a) Show that the intersection

$$U \cap V = \{v \in \mathbb{R}^n \mid v \in U \text{ and } v \in V\}$$

is a linear subspace.

(b) Show by example that the union

$$U \cup V = \{v \in \mathbb{R}^n \mid v \in U \text{ or } v \in V\}$$

may not be a linear subspace.

[1] Hint: in parts (a) and (b), it helps to consider the cases $a \neq 0$ and $a = 0$ separately.

Problem 7. In class, we saw four ways of representing linear subspaces of $\mathbb{R}^n$:

1. The span of a collection of vectors: $\operatorname{span}\{v_1, \ldots, v_r\}$.
2. The image of a matrix: $\operatorname{im}(A)$.
3. The solution set of a homogeneous linear system:

$$\text{the set of all vectors } \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \text{ such that } \begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = 0 \\ \vdots \\ a_{r1}x_1 + \cdots + a_{rn}x_n = 0 \end{cases}$$

4. The kernel of a matrix: $\ker(A)$, the set of all vectors v such that $Av = 0$.

In each part, you are given a representation of a subspace in one of these forms. Find the the three other representations of it.

(a)

$$\operatorname{span}\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

(b)

$$\ker \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 5 \end{bmatrix}$$

(c) The solution set of the system of equations

$$\begin{cases} x_1 - 2x_2 - x_3 + 5x_4 = 0 \\ 3x_1 + x_2 + x_3 + 3_4 = 0 \end{cases}$$

(d)

$$\operatorname{im} \begin{bmatrix} 1 & -1 & 2 \\ 3 & -3 & 6 \\ 1 & -1 & 2 \end{bmatrix}$$